FULL STACK DEVELOPMENT PROJECT

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. INTRODUCTION

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

Team Members

Team Leader: Santha Lakshmi B
Team Members: Hemamalini B, Varshini L, Janani M, Kanishka N

2. PROJECT OVERVIEW

Purpose

The project provides dynamic record-level access control. It checks a selected field value before allowing a user to view or modify a record. The goal is to protect restricted records through automatic, field-based authorization.

Features

• Script-controlled record access
• Field-value based authorization
• Allow or deny access dynamically
• Centralized access-control logic
• Support for testing permitted and restricted records

3. ARCHITECTURE

Frontend

The frontend provides the user interface through which users request or interact with records. It displays the appropriate response after the access-control decision.

Backend

The backend receives record requests and applies the ACL logic. A server-side script evaluates the controlling field and determines whether the requested operation is permitted.

Database

The database stores application records and the field used for access control. The access-control script reads the required field value while evaluating the request.

4. SETUP INSTRUCTIONS

Prerequisites

Required environment depends on the selected implementation. A suitable application runtime, database, scripting environment, and the configured ACL/access-control platform are required.

Installation

1. Obtain the project source code.
2. Configure the application/database environment.
3. Create the required records and controlling field.
4. Configure the ACL rule.
5. Add the field-value validation script.
6. Test permitted and restricted access cases.

5. FOLDER STRUCTURE

Project Structure

Organize the project into frontend, backend, configuration, access-control scripts, and testing sections as applicable to the implementation. Keep ACL scripts and related configuration clearly separated for easy maintenance.

6. RUNNING THE APPLICATION

Application Execution

Start the configured application environment and database. Log in with test users and access records with different field values. Verify that the ACL script allows permitted records and blocks restricted records.

7. API DOCUMENTATION

API Endpoints

Document the backend endpoints used to request, retrieve, update, or manage records. For each endpoint, specify the HTTP method, parameters, access requirement, and expected response.

Example Response

Allowed request: record data is returned. Restricted request: an access-denied response is returned.

8. AUTHENTICATION

Authentication and Authorization

Authentication identifies the user requesting access. Authorization is enforced through the ACL. After a record request, the access-control script checks the selected field value and decides whether the user can perform the requested operation.

Access Control Method

The main authorization mechanism is the Script-Controlled ACL, which dynamically evaluates record data rather than relying only on a fixed permission.

9. USER INTERFACE

UI Description

Provide screenshots of the login/user interface, record view, access-control configuration, and the results for both allowed and denied records.

10. TESTING

Testing Strategy

Test records with allowed field values, restricted field values, empty values, unexpected values, and different user roles where applicable. Confirm that unauthorized records cannot be accessed.

Testing Tools

Use the project's configured development environment, API testing tools, browser testing, and application logs as applicable.

11. SCREENSHOTS OR DEMO

Screenshots / Demo

Include screenshots showing the ACL configuration, script, sample records, successful access, and denied access. Add the project demo or repository link if available.

12. KNOWN ISSUES

Known Issues

Potential issues include incorrect field values, scripting errors, configuration mistakes, and performance impact from unnecessarily complex access-control logic. These should be recorded and addressed during testing.

13. FUTURE ENHANCEMENTS

Future Enhancements

- Support multiple field conditions
- Add role-based and field-based combined authorization
- Add access logs and audit reports
- Provide an administrator dashboard
- Add automated ACL testing
- Improve reusable security-rule management